NTNU

Department of computer science

Examination paper for TDT4175 Information Systems

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Examination date: 27-11-2023

Examination time (from-to): 9:00-12:00

Permitted examination support material: *Allowed aids: C Specified printed and handwritten material is allowed. In the run of the course the book* "BPMN Method & Style", Bruce Silver. (Edition 1 or 2) has been used. Clean copies of the chapters 1-6 used in the course or of the summary text put out on Blackboard are also allowed. *In addition standard dictionaries (English-English, and from English to other languages) are allowed to bring. No additional printed or handwritten materials are allowed. An approved simple calculator is allowed.*

Language: English

Number of pages (front page excluded): 3

Number of pages enclosed: 4

Informasjon om trykking av eksamensoppgave

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Checked by:

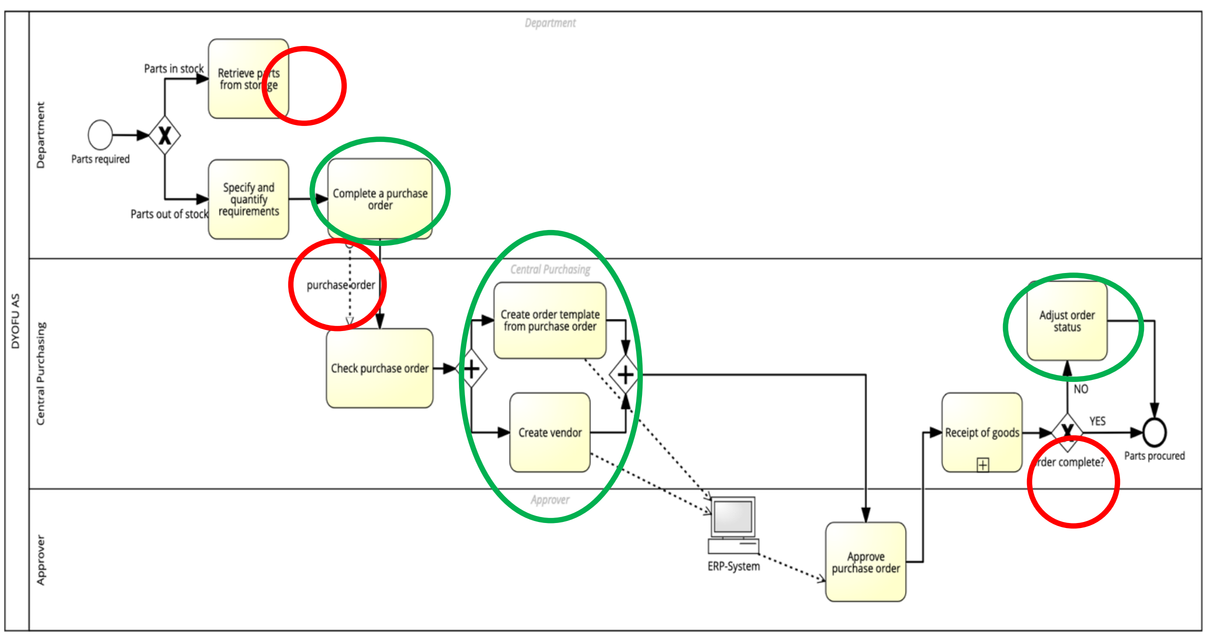
John Krogstie

Date Signature

19/11-2023 John Krogstie

Task 1 - (25 points)
The company DYOFU (Design Your Own FUrniture) produces three different types of furniture: sofa's, dining tables and clothing cupboards. Using a web application customers can configure/tailor the piece of furniture they want to order to a great extent. For example, when ordering a sofa, customers can select a shape from a limited set of possible shapes, the material used for the frame, the size, and the texture and color of the cover. DYOFU is divided into three core departments (Sofa, Table, Cupboard). Because the required materials for the production of ordered furniture cannot be fully predicted, it is essential that DYOFU implements a smooth procurement process. So, when a department finds out that it needs parts/material that is not in stock, it must complete a purchase order and register it in the company's ERP system. This order is processed by the general purchasing department. It first checks the order, if (some of the) ordered items come from an unknown vendor, the vendor is added to the ERP system. Then a so-called order template is generated, consisting of all the separate orders for different vendors. This is also registered in the ERP system and approved by a person that has the mandate to approve orders. This is either the CEO or the head of the finance department. The delivered goods are received by the central purchasing department. If due to delivery problems (part of the) order is not delivered, the order status is adjusted in the ERP system.

Given this case description a student has made the following model.



1.1 (5 points)
The first step of modelling a business process, as advocated by Silver, is choosing the scope of your model. One of the questions to be answered in scoping is: **What does each instance of the process represent?** Answer this question for the model made by the student.

An instance represents the process of ordering (a list of) parts by a department within DYOFU. It starts with the identification of required parts and ends with the check of the received parts.

1.2 (20 points)
The model contains several mistakes, both syntactical and semantical. Find at least two

syntactical and two semantical mistakes (so, four mistakes in total). Motivate why you consider these to be mistakes, and which type (syntactical or semantical) they are.

The red circles indicate syntactical mistakes: activity with no outgoing control flow, use of a message flow within a white box, and doing a check in a gateway. The green circles indicate semantical mistakes. The one in the middle is a wrong use of an AND-gateway. Correct would be to have a check activity 'New vendor', followed by a XOR-gateway. If 'Yes' the control flow goes to the activity 'Create Vendor' and after that to the activity 'Create order template from purchase order'. If 'No' the control flow goes from the XOR-gateway directly to the activity 'Create order template from purchase order'. The two other green circles indicate the omission of a connection to the ERP system while it is explicitly stated that this system is updated. A last semantical mistake, that is more subtle, is related to the end of the process. It would be natural to also include the payment for the received goods in the model, even though this is not explicitly mentioned in the description. In that case we would have a genuine order-to-cash process.

Task 2 (10 points)
Time from order to delivery has always been a weak point for DYOFU. It has been a source of customer complaints and had also weakened DYOFU's position among competitors. It is decided to hire a Business Process Management consultant to find out what are the root causes for this problem. The consultant uses a well-known process mining tool in his analysis.

2.1 (5 points)
What is the data structure called that you need to have to do process mining?

Event log

2.2 (5 points)
What is the process mining activity called where you use the data structure from 2.1 to get insight into how a business process has been executed?

Process discovery

Task 3 (15 points)
Process analysis shows that there are two main causes or bottle necks in the process. The first is the procurement of required materials. This takes often too much time. Sometimes a purchase order is stuck at the central purchasing department before it is sent to the approver. Sometimes it takes a long time before it is approved. And sometimes there is a delay between the approval and the receipt of goods. There are also often delivery problems due to mistakes in the purchase orders. It is decided to improve the workflow between DYOFU and its suppliers by implementing a new information system.

3.1 (5 points)
Which of the information systems discussed in the course is appropriate here? (Not an ERP, that is already implemented).

Supply Chain Management system (SCM)

3.2 (10 points)
Describe how this system can help to improve the workflow between DYOFU and its suppliers.

A SCM is characterized as:

Supply chain management (SCM) is a system that includes planning, executing, and controlling all activities involved in:

- ✓ Sourcing and procurement of raw materials
- ✓ Converting raw materials to finished products
- ✓ Warehousing and delivering finished product to customers

SCM manages materials, information, and finances as they move from:

- Supplier -> Manufacturer -> Wholesaler -> Retailer -> Consumer

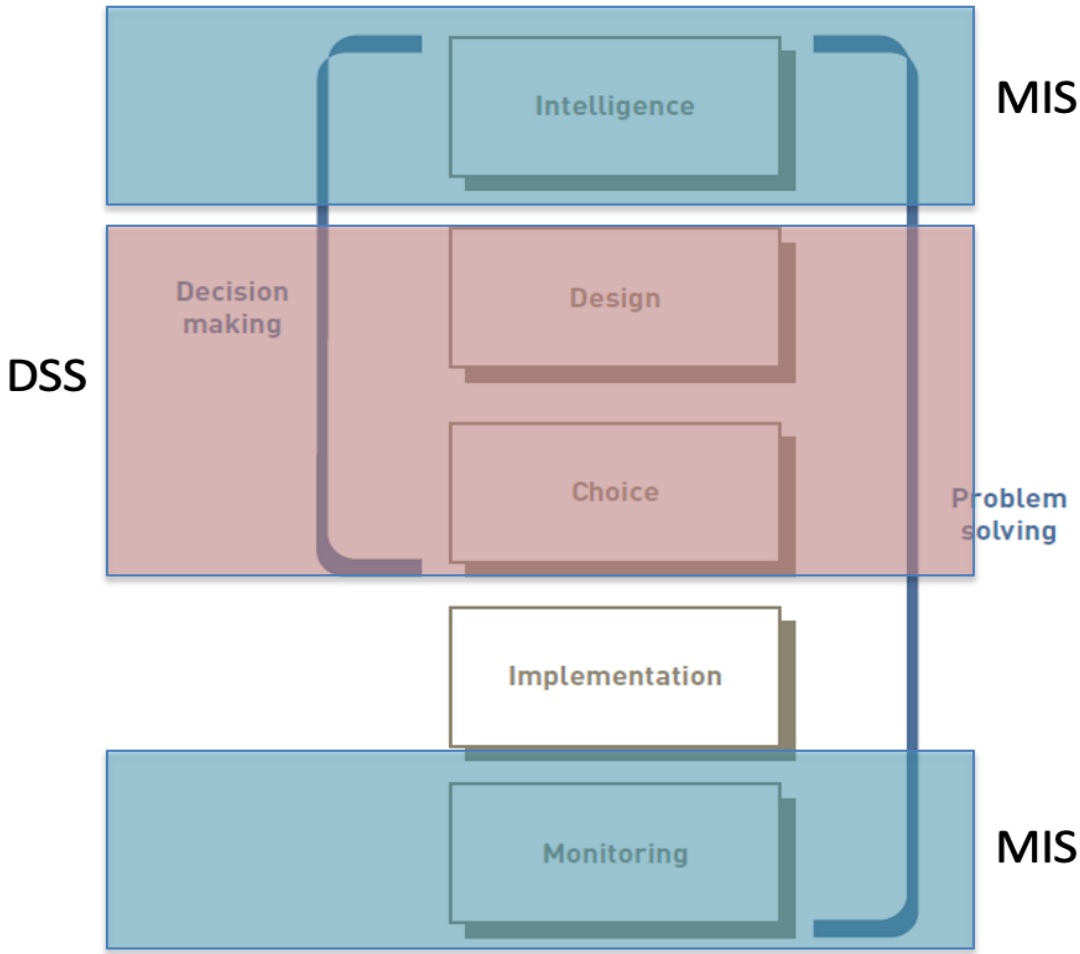
In our case the focus should be on 'Sourcing and procurement of raw materials', Supplier -> Manufacturer. This could be supported by a B-to-B e-commerce system, enabling direct ordering to the right vendor by a department after approval. So, bypassing the central purchasing department. Or, more advanced, giving vendors insight in the order portfolio, demand planning and stock inventory. This enables them to send in an offer for the required materials. Other answers are also possible.

Task 4 (25 points)

The second problem that was identified in the process analysis was that the workload for the different core departments was often quite uneven. So, sometimes Sofa had a lot of orders to process, leading to a production queue, while Table had too few orders. Historical data analysis showed that there also were seasonal variations: more tables in September/October and more Cupboards in April/May. It was decided to re-organize DYOFU to get a more balanced workload. Instead of working with one product all employees will now work with all three. So, depending on the order portfolio, one could work on a sofa, a table or a cupboard. This requires good people planning.

4.1 (10 points)

People or workforce planning is a typical example of informed problem solving. In the lecture on business information systems, Part 2, we discussed a model for informed problem solving. Describe the five steps distinguished in this model.



4.2 (5 points)
Management Information Systems (MIS) and Decision Support Systems (DSS) were discussed as IS types that can support informed problem solving. Define a DSS and describe the types of decisions a DSS can support to make.

- DSS:
  - Organized collection of people, procedures, software, databases, and devices used to help make decisions that solve problems
- Focus of a DSS:
  - Is on decision-making effectiveness regarding unstructured or semistructured business problems

4.3 (10 points)
Describe how a MIS and a DSS can help in the workforce planning in our example.

First the MIS. In the first step in the informed problem solving 'Intelligence', a MIS can help in finding out two things:

1. How many orders of which product are there in portfolio for planning period? (And maybe: how many can we expect to come in addition?)

2. How is our personnel resource situation in the planning period? This includes how many people are there, who is on holiday, who is on sick leave, expected sick leave percentage in the planning period, etc.

Given these insights a DSS can help in finding a good resource allocation, accommodating the required flexibility to handle unplanned sick leave and rush orders.

Finally, a MIS can support 'monitoring'. When new rush orders come in, our unplanned sick leaves occur that disturb the resource planning, this should be noted. Also, when delivery problems occur, causing delays in the production.

Task 5 (15 points)
Removing the three core departments and as a result, the specialization of personnel in either sofa, table or cupboard construction, is a major organizational change. Vi have discussed two ways of doing this: ***reengineering*** and ***continuous improvement***.

5.1 (5 points)
What is the difference between these two ways of realizing organizational change?

Reengineering
- Also called process redesign and business process reengineering (BPR)
- Involves the radical redesign of business processes, organizational structures, information systems, and values of the organization to achieve a breakthrough in business results

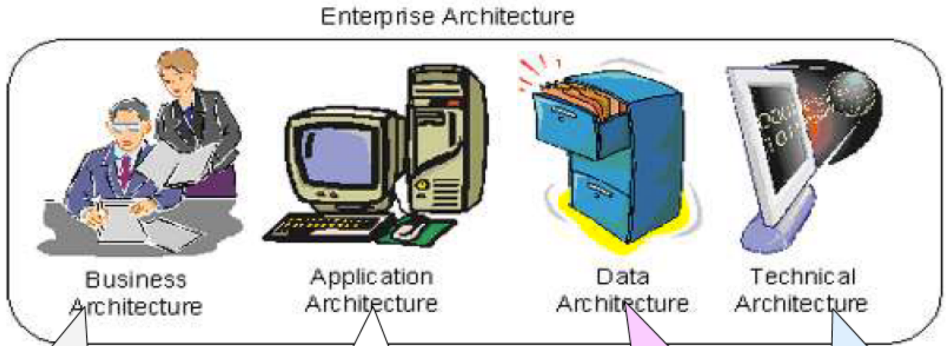
Continuous improvement
- Constantly seeking ways to improve business processes and add value to products and services

5.2 (10 points)
Which of these two strategies would you consider to be most appropriate in the DYOFU case?

The most appropriate way of executing the change here, would be reengineering. The root cause for the problem with long delivery times is the silo organization: every department tries to optimize its own production process. Resource planning is local (per department) and also coping with variations.

Task 6 (10 points)
The number of information systems implemented in DYOFU is quite considerable. They have an ERP system, a design tool, a web portal, a MIS and a DSS among others. Some of these systems are integrated, others not. Management wants to have a better overview of the IS infrastructure and its alignment with business organization and strategy. They decide to hire an Enterprise Architect. She will describe DYOFU's enterprise architecture using TOGAF. Which four architectures are part of such an EA description?

TOGAF’s Enterprise Architecture



Describes the processes the business uses to meet its goals.

Describes how specific applications are designed and how they interact with each other.

Describes how the enterprise datastores are organised and accessed.

Describes the hardware and software infrastructure that supports applications and their interactions.